

4

SYNTAX DIRECTED TRANSACTION

4.1 Syntax Directed Transaction

CFG + Transition } SDT
Syntax + Transition }

SDT: CFG + Transition → 1) Meaning 2) Semantic

4.1.1 Application of SDTL

- (1) Used to perform Semantic Analysis.
- (2) Produce Parse Tree.
- (3) Produce intermediate representation.
- (4) Evaluate an expression.
- (5) Convert infix to prefix or postfix.

Example: $S \rightarrow S_1 S_2$ [$S.count = S_1.count + S_2.count$]

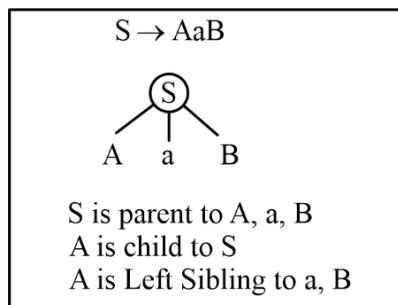
$S \rightarrow (S_1)$ [$S.count = S_1.count + 1$]

$S \rightarrow \epsilon$ [$S.count = 0$]

- Count is an attribute for non-terminal

4.1.2 Attributes

- (1) Inherited Attribute
- (2) Synthesized Attribute



- (1) Inherited Attribute: (RHS)

$S \rightarrow \boxed{A} B \{A.x = f(B.x \mid S.x)\}$

- The computation at any node (non-terminal) depends on parent or siblings (S).

- In above Example, x is inherited attribute.

(2) Synthesized Attribute: (LHS)

$$S \rightarrow AB \{S.x = f(A.x \mid B.x)\}$$

x is synthesized attribute.

The computation at any node (non-terminal) depends on children.

4.1.3 Identifying Attribute Type

- Always check every Translation.

- | | | |
|---|---|--|
| 1) $D \rightarrow T : L ; \{L.Type = T.Type\}$ inherited
$L \rightarrow L1, id ; \{L1.Type = L.Type\}$ inherited
$L \rightarrow id$
$T \rightarrow integer \{T.Type = int\}$ synthesized | } | Type is neither inherited nor synthesized |
| 2) $E \rightarrow E1 + T \{E.val = E1.val + T.val\}$ synthesized
$E \rightarrow T \{E.val = T.val\}$ Synthesized
$T \rightarrow T1 - F \{T.val = T1.val - F.val\}$ Synthesized
$F \rightarrow id \{F.val = id.val\}$ synthesized | } | Val is synthesized attribute |
| 3) $S \rightarrow AB \{Aa = B.x; S.y = A.x\}$ x is inherited y is synthesized
$A \rightarrow a \{A.y = a\}$ y is synthesized
$B \rightarrow b \{B.y = a.y\}$ y is synthesized | } | $x \Rightarrow$ inherited
$y \Rightarrow$ synthesized |
| 4) $D \rightarrow T L \{L.in = T.type\}$ inherited(in)
$T \rightarrow int \{T.type = int\}$ /synthesized
$L \rightarrow id \{Add\ type(id.entry, L.in)\}$ | } | $in \Rightarrow$ inherited
$type \Rightarrow$ synthesized |

4.1.4 Syntax Directed Definitions (SDDs): (Attribute Grammar)

(1) L-Attributed Grammar

- Attribute is synthesized or restricted inherited. (1) Parent 2) Left sibling only).
- Translation can be appended anywhere is RHS of production.
- Example: $S \rightarrow AB \{A.x = S.x + 2\}$
 or, $S \rightarrow AB \{B.x = f(A.x \mid S.x)\}$
 or, $S \rightarrow AB \{S.x = f(A.x \mid B.x)\}$
- Evaluation: In Order (Topological).

(2) S-Attributed Grammar:

- Attribute is synthesized only.
- The transaction is placed only at the end of production.
- Example: $S \rightarrow AB \{S.x = f(Ax \mid Bx)\}$.
- **Evaluation:** Reverse RMD (Bottom-Up Parsing).

4.1.5 Identify SDD

- (1) $E \rightarrow E_1 + E_2$ {E.type = if (E₁.type == int & & E.type == int) then int} Synthesizer else type – error.
 $E \rightarrow id$ {E.type = Lookup (id.entry)} synthesizer.
- type is synthesized, hence S-attribute and also L-attributed Grammar.
 - Every S-attributed Grammar is also L-attributed Grammar.
 - For L-attributed Evaluation, use the In-order of annotated Parser Tree.
 - For S-attributed, reverse of RMD is used.
- find RMD Order.
 → Consider its reverse.

